Chapter 4
Anatomy of a software house

The purpose of this chapter is to study in detail a hypothetical company in order to understand the application, in practice, of some of the ideas described in the two previous chapters. Although the company is hypothetical, it is based on the author's experiences in the software industry.

4.1 The company

Syniad Software Ltd was founded some ten years ago by four friends and colleagues who still own most of the shares in the company; all four are members of the Board of Directors, along with two others who were recruited later.

The company specializes in the production of bespoke software for clients who demand work of high quality; much of this work is advanced, in the sense that it demands the use of techniques that have only recently been developed or, in some cases, demands the development of completely new techniques. Along with this development work, there is a significant amount of consultancy work in similar areas.

Syniad's head office is some 40 miles west of London. An office south of Manchester was opened in 1993 and some 40 staff are based there. An office was opened in Delft in the Netherlands in 1996; the only staff based there are its manager, a salesman and a secretary but a number of contracts have been won and carried out using UK-based staff.

4.2 Company structure

The top level structure of the company is shown in Figure 4.1.

The Operations Director is responsible for all the revenue earning operations of the company. It is his job to ensure that all projects are completed satisfactorily and that resources are available to carry out the projects that the company wins; he is also responsible for ensuring that the utilization of

Figure 4.1 Syniad's top management structure.

revenue earning staff is kept at a satisfactory level. As is natural, given these responsibilities, the personnel function reports to him. The Technical Director is responsible for:

- quality management;
- research and development;
- marketing at a technical level (e.g. arranging for staff to give papers at conferences);
- technical training (as opposed to training in, say, project management or presentational skills, which are the responsibility of the personnel function).

Some aspects of this structure deserve comment.

The separation of responsibility for quality management from responsibility for day-to-day operations is much to be commended; as we have said in Chapter 2, such separation is essential for effective quality control. The division of responsibility for training could potentially lead to conflict, as could the division of responsibility for marketing. In practice, there are few problems because the individuals in question have good personal relationships with each other and take care to discuss all areas of possible conflict.

More problems arise from the position of the personnel function. There is, inevitably, a tension between the Operations Director's desire to maximize staff utilization and revenue, on the one hand, and the individual employee's career aspirations on the other. A specialist in IBM mainframe software may feel, for example, that career opportunities in that field are limited and wish to gain experience of other types of hardware; however, so long as the company has projects which involve IBM mainframes, such a specialist can command a higher charging rate when used on such work and it will therefore be in the interests of the Operations Director not to provide the opportunity to move to other work. Similar problems arise when the Operations Director wants an individual to work abroad but the individual's personal circumstances make this inconvenient or undesirable. The personnel function should be able to act as an intermediary in these circumstances but it is difficult for it to do so while it reports directly to the Operations Director.

A similar tension may also exist between the Operations Director's short-term objectives and the longer term goals of the company. It is sometimes said that the art of running a software house is the art of matching people to projects in an optimal way; what is optimal depends, of course, on individuals' aspirations and the organization's strategic plans, as well as short-term needs.

The groups are each managed by a group manager; a group consists of somewhere between 15 and 50 staff working in a reasonably homogeneous area. Some groups serve specific market sectors, such as banking, while others (usually the smaller ones) cover specific technical specialities such as databases. When potential new business is identified, it is assigned to a specific group. This group will then be responsible for providing the sales staff with the support necessary to win the business and for carrying out the contract when it is won. In theory at least, staff have a sense of belonging to a group and regard their group manager as the manager who is permanently responsible for their career in the company. In practice, because projects often require expertise from more than one group, staff often find themselves working on projects for groups other than the one to which they belong and the sense of group identity is diluted.

This points up what is perhaps the most serious problem of organizational structure which Syniad faces, what we may term “clashing structures”. Consider what happens if the company is asked to bid for installing a communications system for a bank in the Netherlands. The manager of the Dutch office will have made the initial contact and has local knowledge which will be essential if the contract is to be won. However, in order to put together a credible bid, he will need assistance from specialists in the market sector (banking) and in the technology (communications); he may well need assistance also, perhaps in the form of copies of proposals from similar bids in the past, from the sales and marketing staff at Head Office. He himself reports principally to the Overseas Director but has to keep the Operations Director satisfied about the running of projects and the staff utilization, the Sales and Marketing Director happy about sales and the Finance Director content with the way in which the Dutch company’s finances are handled. Inevitably, this gives rise to conflicts over priorities.

It is worth remarking that Syniad’s organizational structure shows elements of all three of the types of organizational structure described in Section 2.3. At board level, there is a basically a functional division of responsibilities but with a geographical element represented by the director respon-sible for overseas operations. Within the operations division, the group structure is analogous to a product line organization in a manufacturing company. In a company of the size of Syniad, the distinction between centralization and decentralization has little meaning. Centralized policies and procedures are widely used but they have usually been developed within one part of the company and have been adopted by general consent.

4.3 Management of staff

Management of staff has been critical to Syniad’s success so far and will remain so in the future. The company’s ability to expand depends on its ability to recruit and retain staff; its reputation in the market place depends on the quality of its staff. The directors and senior management are well aware of this and much time and effort is devoted to it.

When the company was small, with less than about 50 staff, there were few problems. The directors knew all the staff personally and most had been recruited through personal contacts. Staff were loyal to the company and this loyalty showed itself in a willingness to work unpaid overtime when necessary and to maintain high technical standards regardless of the extent of supervision. For their part, the directors put a lot of effort into keeping all employees informed about the company’s activities and were able to take a sympathetic attitude to any personal problems that arose. Although the company was not able to organize much training itself, it encouraged and was willing to pay for staff to go on training courses provided by other organizations. Almost no one left the company during this period.

As the company grew, problems became apparent. Although the general quality of the staff remained high, it was inevitable that some recruits proved to be less able and less industrious than expected. The directors were not able to build up the same level of personal relationships with the new staff and loyalty to the company diminished. Because of the high quality of the original staff, many of them were now in senior positions; this,

together with their long-standing personal ties with the directors, led newer staff to feel that the company was run by a clique and that they would always remain outsiders. Although this feeling was patently contrary to the facts—many senior staff were recent recruits—it was not conducive to building up company loyalty. The rate at which staff were leaving the company became a cause for concern.

Management reacted swiftly to these problems. Staff appraisal procedures were set up with the object of ensuring that:

- employees' achievements and contributions to the company were properly recorded;
- staff knew what was expected of them and what they needed to achieve in order to gain promotion;
- proper plans for training and career development were made and regularly reviewed;
- employees were aware of the company's opinion of their performance.

Employees are appraised and their salary reviewed every six months. Appraisals are normally carried out by the manager to whom the employee is directly responsible; they are then reviewed by the employee's group manager. At the same time as the staff appraisal scheme was introduced, the group structure was created and group managers were specifically charged with building up good personal relationships with their staff and with keeping staff fully informed about Syniad's activities. The result of these changes was that morale increased again and the rate of loss of staff decreased, although the atmosphere of the company when it was small was never recaptured.

Despite the overall success of Syniad's staff management, there are still perceived to be a number of difficult problems to be overcome:

- how to provide continuity of management as staff move from project to project;
- how to exploit and foster group expertise (e.g. the collective expertise of the company's six database experts) in an environment in which the members of the group are usually working on different projects;
- how to maintain standards and company loyalty as the company continues to grow.

The biggest difficulty, however, in developing and pursuing a consistent and satisfactory personnel policy, is the unstable nature of the business. Syniad lurches from the position in which it is desperate to recruit staff to service the contracts which it has, to the position in which it is desperate to get business in order to keep its staff on revenue earning work. In either of these situations it is difficult to ensure that staff work on projects which are appropriate to their skills, their desires and their planned career development. This is ultimately a consequence of the company's under-capitalization, a point that we shall develop further in Section 4.6.

4.4 Producing the budget

Staff in the company are broadly divided into technical or *revenue earning* staff and *non-revenue earning* staff; the latter include management, secretarial, accounting and so on. The crux of the problem in producing the budget is to estimate the costs of the revenue earning staff and the revenue which they can be expected to earn. Overhead costs,

including the costs of non-revenue earning staff must then be contained within the sums likely to be available.

Revenue earning staff are divided into seven grades, numbered 1 to 7, with grade 7 as the most senior. For each grade, the average direct cost (i.e. salary +National Insurance+employer's pension contribution+perks) in 1999 was calculated, using the July 1999 payroll. The average number of revenue earning days per employee in each grade was calculated for the period 1 August 1998 to 31 July 1999; this number varies from grade to grade because:

- different grades have different training needs;
- senior grades tend to spend more time on non-revenue earning activities such as sales support;
- senior technical staff are usually more specialized and it therefore takes longer to redeploy them once they have completed a project.

The figures estimated for 1999 are shown in Table 4.1.

The 2000 figures have been prepared using the 1999 figures as a base, adjusting them in the light of the following inputs:

- the sales staff's estimates of how much more business they could sell, and at what rates, if the staff to handle the business were available;
- the personnel department's estimates of the company's ability to recruit new staff of a suitable calibre and to retain existing staff;
- the personnel department's estimate of salary trends in the industry;
- the operations director's view of the staff utilization rates achievable;
- specific expansion plans, e.g. plans to open an office in a new location;
- the financial director's estimate of the company's ability to raise funds for expansion.

The 2000 figures are shown in Table 4.2.

The work of preparing a budget needs to be started well in advance of the period to which it refers and so estimates have to be made on the basis of the best information available. The figures for the first half of 1999 have therefore been used to estimate the figures for the whole of 2000. If the second half of 1999 shows a marked difference from the estimates, it may prove necessary to revise this table, and hence the whole budget, at a very late stage. While this may be unavoidable, it is certainly undesirable. If the revenue is lower than estimated, it will upset plans and commitments already entered into. Nothing is worse for the morale of the staff than to be forced to withdraw from plans that have already been announced. Even if the position is better than predicted, the result may be a scramble to implement ill prepared and badly thought out plans for expansion.

The uncertainty about the direct costs and revenues for 2000 is inherent in the budgeting process. More avoidable is the excessively optimistic character of the predictions. What is being predicted is an increase in staff of about 16 per cent; an increase in the already high level of staff utilization; and a substantial increase in charging rates. The author's experience suggests that this combination of performance improvements will only be possible in a very

Table 4.1 1999 staff costs and revenue figures (£).

Grade	*Number*	*Annual cost*	*Days worked*	*Utilization (percent)*	*Daily cost*	*Charge rate*	*Total cost*	*Total revenue*
1	20	18,500	195	87	94.87	195	370,000	760,500
2	25	23,000	205	92	112.20	220	575,000	1,127,500
3	42	27,000	206	92	131.07	255	1,134,000	2,206,260
4	41	30,500	204	91	149.51	305	1,250,500	2,551,020
5	27	36,600	198	88	184.85	370	988,200	1,978,020
6	20	39,800	185	83	215.14	440	796,000	1,628,000
7	15	44,000	160	71	275.00	560	660,000	1,344,000
Totals	190			88(avg)			5,773,700	11,595,300

Table 4.2 Predicted staff costs and revenue figures for 2000 (£).

Grade	*Number*	*Annual cost*	*Days worked*	*Utilization (percent)*	*Daily cost*	*Charge rate*	*Total cost*	*Total revenue*
1	25	19,500	198	88	98.48	215	487,500	1,064,250
2	30	24,150	208	93	116.11	245	724,500	1,528,800
3	50	28,350	210	94	135.00	290	1,417,500	3,045,000
4	50	32,025	207	92	154.71	350	1,601,250	3,622,500
5	32	38,430	200	89	192.15	410	1,229,760	2,624,000
6	24	41,780	190	85	219.89	500	1,002,720	2,280,000
7	19	46,200	165	74	280.00	630	877,800	1,975,050
Totals	230			90 (avg)			7,341,030	16,139,600

well run company, blessed with excellent sales staff, in the most favourable of market conditions. The 2000 figures might be acceptable as performance targets but should not be used as a basis for planning.

The other side of the budgeting process concerns overhead costs, that is costs which are not directly related to the company's ability to generate revenue [1] . Table 4.3 shows the 1999 estimates and the predicted figures for 2000. The general principle used in arriving at these figure has been to increase all the 1999 figures by 5 per cent to allow for inflation and then to increase by a further 21 per cent those overheads that are expected to increase proportionately to the planned increase in staff. The two main exceptions are rent and rates, and interest. In the case of rent and rates, the company's rents are fixed

and not due for review in 2000; the only increase is therefore that predicted for the business rate.

Interest is more complicated. A separate cash flow prediction is produced from which the cash needs of the company in 2000 can be established. Once it is decided how to finance these, the interest payments can be calculated. As a rule of thumb, a company such as Syniad needs working capital of about 25 per cent of its annual turnover, i.e. about £3 million for 1999. Depending on

Table 4.3 Overheads for 1999 and 2000 (£).

	1999	*2000*
Rent and rates	700,000	720,000
Building maintenance	84,000	88,200
Heating and lighting etc.	40,000	42,000
Cleaning	20,000	21,000
Telephone and postage	85,000	108,039
Equipment depreciation	20,000	25,421
Equipment maintenance	12,000	15,253
Directors' salaries and expenses	480,000	504,000
Management salaries and expenses	945,000	1,201,145
Sales and marketing salaries and expenses	900,000	1,143,947
Secretarial	190,000	241,500
Library and information services	32,000	40,674
Training courses	180,000	228,789
Insurance	25,000	31,776
Legal expenses	20,000	25,421
Interest	400,000	600,000
Total	4,133,000	5,037,166

1. A separate, and comparatively small, budget is produced for capital expenditure and it is from this that the figure for depreciation is obtained. Company cars are leased and do not therefore give rise to any depreciation.

interest rates, this can lead to an interest charge of anything from £350,000 to £750,000; we have assumed figures of £400,000 for 1999 and £600,000 for 2000.

The budgeted figures for 1999 and 2000 look very healthy for this type of company. The predicted profit for 1999 is £1,688,600 which is 15 per cent of turnover; for 2000 it is £3,761,404, 23 per cent. However, a 10 per cent drop in turnover—perhaps caused by

failure to maintain the level of the charging rates or by a drop in staff utilization—would reduce the 1999 profit to £532,070, only 4.6 per cent. Moreover, a worst case scenario is the one in which the costs, both direct and indirect, for 2000 attain their budgeted level, £12,378,196, while revenue remains at the 1999 figure of £11,595,300, to give a loss of £782,896.

This demonstrates very clearly that the company's expenditure plans for 2000 cannot be allowed to go ahead in full, all at once, but must be conditional on the revenue and forward sales figures showing the right trend. In other words, the company's financial performance must be carefully monitored.

4.5 Monitoring financial performance

Monitoring Syniad's performance against the budget should, in principle, be straightforward. Each month, the income and expenditure under the various heads are compared and, if significant deviations are observed, corrective action is taken. In practice, this simple procedure presents many difficulties.

4.5.1 Costs and revenue

A major problem is caused by random fluctuations, themselves the product of many individual factors, for example:

- The budget assumes that, over the twelve month period, staff will increase by a total of 30; in other words the excess of joiners over leavers will be 30. In the absence of any reason for believing otherwise, this will have been shown as a net monthly increase of 2.5 staff. However, while the budgeted figure of 30 may well prove to be correct, it is unlikely to follow this neat pattern; in some months there may be a net increase of seven or eight, while in other months there may be a net decrease.
- Large projects will cause significant deviations from the budget in various ways. The call for proposals for a large project will take several senior technical staff away from revenue earning work on to the preparation of the proposal; this may last for as much as a couple of months, or even longer. On average over the year, the utilization of that grade of staff may accord with the budget but it will be much lower for the period of proposal preparation. During a big project, the overall level of staff utilization may increase, because so many staff are continuously and fully employed on the project but the end of the project often sees staff becom-ing available at a rate greater than that at which they can be deployed on to other projects, thus adversely affecting utilization.
- In fixed price projects, revenue is accrued on the basis of the increase in the value of work in progress. Even the most conscientious of project managers will not normally re-estimate the whole project every month; rather, the increase in work in progress will be calculated monthly by applying a formula mechanically and, once every few months, the project will be re-estimated. The result of this is likely to be a hiccough (one way or the other) in the monthly revenue—it may even turn out to be negative.

4.5.2 Project costing

Because of these difficulties in monitoring the overall performance of the company, Syniad also tries to monitor the financial performance of individual projects, through a project costing system. The costs and revenue of each project are calculated each month and the cumulative gross margin (i.e. the difference between total costs and total revenue to date on the project) calculated as a percentage of the total revenue. Project managers are set a target of 60 per cent for the gross margin on their projects.

In practice, this system does not work well. Project managers complain that there is usually little they can do about the costs, since these are dominated by the payroll costs of the staff, and nothing they can do about the rates, since these were determined by the sales staff who carried out the initial negotiations. In addition, they usually have difficulty reconciling their records of the non-staff costs with those used by the accounts department, because of uncertainty about which month a charge will be allocated to.

4.5.3 Sales

The budgeted increase in revenue derives partly from increased charge rates, partly from better staff utilization and partly from an increased number of staff. All these factors are influenced by the forward sales position, that is by the staff required and the rates earned on the work to which the company is committed in the coming months. Two reports are used for assessing and monitoring the sales position.

The *confirmed sales report* shows, for each grade, the number of staff in that grade who are committed to contracts in each of the following twelve months and the total expected revenue from that grade in each month.

The *sales prospects report* shows, for each sales prospect, the potential value of the sale, its likelihood and the likely start date. The likelihood is reported on the following scale:

- 0.1 indicates that the client has an identified requirement which is expected to be met by commissioning a software house, but that Syniad has not yet been asked to quote for the business and there is no certainty of being asked;
- 0.3 indicates that Syniad has been asked to quote for the business or will be asked in the near future;
- 0.5 means that Syniad has quoted for the business and is short listed or, for other reasons, is in competition with no more than two other companies;
- 0.7 means that Syniad has been offered the business, subject to negotiation (possibly on price, possibly on contractual conditions, possibly on technical content);
- 0.9 means that negotiations are concluded but no signed contract has yet been received.

Prospects that merit a likelihood of less than 0.1 are not included in the sales prospects report. These likelihoods are based on experience which suggests, for example, that Syniad will win, on average, about three out of ten contracts for which it has been asked to bid. Obviously, much of the information in this report is very subjective but the report still provides a useful and disciplined way of assessing the fruits of the salesmen's activities.

Taken together, the confirmed sales report and the sales prospects report enable a reasonably accurate assessment of future staff needs and revenue to be made. In particular, they allow recruitment tactics for the next two or three months to be determined. Note, however, that this is at a very coarse level; a need for an increased number of grade 4 staff may be apparent but the reports do not indicate, for example, whether they should have experience in air traffic control systems or payroll.

A particular difficulty arises with large projects at a likelihood of 0.5. Such a project, if the company is successful in winning it, may suddenly require a fairly substantial number of staff. There is a considerable risk in recruiting the staff before the project is won but it may be impossible to recruit them quickly enough if the company waits until it knows whether it has won the contract. Moreover, it seems to be in the nature of such projects that the decision date is frequently deferred but, the moment the decision is taken, the client expects an immediate start.

4.6 Long-term planning

As well as short-term, i.e. year-by-year plans, a successful company needs to have strategic plans covering, say, the next five years. The day-to-day hassle of running the company makes it very difficult to produce strategic plans and, even more, to monitor the progress of the company towards their achieve-ment. Nevertheless, the ability to plan strategically and to achieve strategic objectives is the hallmark of well run, successful companies.

Strategic planning in Syniad has two related aspects. The first is to identify appropriate long-term goals and the second is to identify and formulate plans to overcome those problems which are inhibiting it from attaining these goals. We are talking here, of course, of what may be called “structural” problems rather than day-to-day problems.

4.6.1 Expansion plans

The directors of Syniad want to see the company become one of the leading companies in its field in Europe. Although it is difficult to estimate the size of the market for the type of services which it provides (because publicly available figures aggregate its market with many markets in which it has no interest), it is clear from the size of the companies which it sees as competitors that, at present, its UK market share is less than 5 per cent and its market share elsewhere in Europe is minuscule. The directors wish to see the company ultimately attain a market share of 15 per cent in the UK and 10 per cent across the rest of Europe. As a step towards this, they are formulating a plan to raise the UK market share to 10 per cent and to attain a market share of 5 per cent in Holland, Germany, Denmark and Italy, within the next five years. It is estimated that this means increasing the staff from the present 200 to around 1200.

Expansion on the scale desired depends on the ability to recruit and retain suitable staff and on the ability to raise the necessary funding. It will also require the ability to run a much larger and multinational organization.

A major strategic decision concerns the staffing of the foreign operations. The model

used so far in Holland has been to employ a nucleus of sales and management locally and to staff the projects from the UK. There are considerable advantages in continuing with this model:

- it retains the idea of a single pool of technical staff available for any project, and thus makes it easier to find appropriate staff for each project and easier to maintain a satisfactory level of utilization;
- the company knows how to manage UK staff but managing foreign staff is bound to be different—employment legislation, for example, varies very widely amongst the countries of the European Community—and it is difficult to find managers with suitable expertise;
- UK based staff cost less than staff employed in most other European countries, even allowing for the need to pay subsistence expenses when they are abroad.

However, despite the success of this model on the comparatively small scale of the Dutch operation, it might very well not work elsewhere. In particular, the Dutch are both able and willing to carry out technical work in English; this will be much less true of Germany and Italy. The proposed expansion would mean that, instead of the present 10 per cent of Syniad's business being carried out on this basis, something like two thirds of the business would be done in this way. This would mean recruiting a very large number of staff prepared to spend a lot of time abroad and many of them would need language skills. This does not seem possible.

The decision has therefore being taken that Holland will continue to operate on the present basis and that the Danish company, when it is established, will operate on the same basis, since Denmark is a comparatively small market in which English is always likely to be acceptable as the working language. The German and Italian companies, however, will operate primarily with local nationals from the start, although specialized technical expertise will be brought in from the UK when necessary.

In the case of Germany and Italy, there remains the question of whether to recruit local managers to start the companies from scratch or whether to try to buy existing companies. Buying an existing company has the advantage that the risks are lower; the company will already have contracts which are generating revenue and will have the contacts to win further contracts in the future. On the other hand, the initial investment will be large and Syniad will have to raise a lot of capital; there may be difficulties in integrating the company with Syniad; and there is the risk that the staff in the company will be unhappy about the change of ownership and leave rapidly, thus destroying the value of the investment. Building a company from scratch involves much less investment and there will be fewer problems of integration but the risks are high; success depends crucially on being able to recruit the right person as local manager. Initially it has been decided to pursue both avenues. A search will be made both for suitable local managers and for suitable companies; the position will be reviewed when more is known about the availability of candidates in either category.

4.6.2 Company image

Syniad already has an image as a software house producing high-quality, state-of-the art

software for demanding and prestigious clients; such clients are usually prepared to pay a slight premium for using Syniad's services on projects which demand high quality. The directors are happy with this image.

There is one respect, however, in which the image of Syniad—and, indeed, the image of all comparable companies—is deficient. Firms of management consultants have succeeded in building an image of providing high value services at a high price; their charging rates are typically some 50 per cent higher than Syniad's, for comparable staff. A significant amount of their work is comparable to that carried out by Syniad. It is particularly galling to see a member of staff, often not one of the most competent, leave Syniad to join a firm of management consultants who then charge for the ex-employee's services at a much higher rate.

It is a long-term aim of the company to bring its charging rates up to the level of management consultants. Apart from exhorting the sales force to negotiate the highest possible rates, however, there is no strategy for bringing this about. If it were achieved, it would enable much more money to be spent on training and information services; it would enable target staff utilization levels to be reduced, thus making it easier to assign staff to projects in accordance with their career development plans; and it would enable the company to invest more in technology for improving the software development process. All these, in turn, would improve still further the company's quality image and assist it in recruiting and retaining staff, so ultimately furthering its expansion plans.

4.6.3 Product mix

At present, Syniad relies for its income entirely on *fee-based revenue,* that is revenue that is tied to fees charged for the services of individual employees. As a result, competition means that its income is limited to some multiple of the payroll cost of the revenue-earning employees; what this multiple is depends upon the company's image and its success in persuading clients of the high quality of its staff, but it is unlikely to rise much higher than about 2.5. Profit, as a percentage of turnover, will depend on the efficiency of the operation, in particular on maintaining high utilization while keeping overheads low. Such efficiency will, however, tend to militate against the high quality that the company is pursuing—maintaining high-quality staff requires significant expenditure on staff training and development, for example.

The directors of Syniad look enviously at the producers of packaged software. Once a company has made the initial investment in developing a package, large numbers of copies may be sold at very little extra cost and without employing large numbers of extra staff. This frees the company from the limitations of fee-based revenue. However, product development is expensive and risky; only a small proportion of the software products developed achieve a level of sales that really justifies the initial investment. Syniad does not have the money to develop new products. Even if it did, it is doubtful whether the directors would be prepared to risk investing it in speculative development. They realize that the company does not have the expertise or the infrastructure to enter the very crowded market for high-volume packaged software. Instead, the policy is to look for opportunities to develop specialized products that build on experience gained in bespoke projects or to seek licensing agreements from overseas suppliers of such

products. To date the policy has shown few signs of success.

4.6.4 Finance

Many of the problems that have been noted are a consequence of the under-capitalization of Syniad, that is, there is not enough money invested in the company; this is a problem common to many similar companies. One difficulty is that the result of increased investment would be too ephemeral to appear on the balance sheet or be used as security for a loan. Some of the capital raised might be spent on hardware, which certainly would appear on the balance sheet. But most of it would be spent on new or improved procedures, standards and methods; on increased training and improved career development for staff (including financing a lower utilisation rate); on improved library and information services; and on product development. While such expenditure would certainly help the company to win more business and charge higher rates, its results are intangible and are valueless outside the context of the company. This means that it must be funded from profits, from loans secured against the personal property of the directors, or from new equity capital; profits and secured loans simply cannot generate the investment required so the company must look for new equity.

The desire to expand into Europe, with the concomitant acquisition plans, also makes it necessary to raise further funding. Again it is doubtful whether this could be done completely from loan capital; the risks involved are such that potential lenders will require a say in the running of the company which only a shareholding will give them.

The company is handicapped by the large interest charges that it is paying for its overdraft, essentially the working capital it needs to cover the gap between invoicing and receiving payment. If this could be funded internally, the company's profit would look much healthier.

If Syniad were to be sold, the purchaser would probably be prepared to pay somewhere between £15 and £20 million. Since the shares are almost all held by the original four founders, they are, on paper, very rich people. However, they have no access to this wealth; they can only live on the salaries they are paid as directors. Inevitably, there will come a time when they wish to turn some, at least, of this wealth into cash. This means selling some of their shares.

All of these factors lead inexorably to one conclusion. Fresh equity capital is needed. It can be obtained only by selling a part or the whole of the company. This may be done by selling the company to another company, or by turning the company into a public company with a stock exchange flotation, or in various other ways. (There are many recent examples of real companies similar to Syniad following one or other of these paths.) However it is done, such a change will affect the character of the company and the working lives of its staff.

4.7 Conclusions

Syniad, despite its problems, is a successful and well-managed company. However, the directors' grandiose expansion plans, if they were to come to pass, would change the

company from a comparatively small, largely uk company into a large (by software house standards) multinational. Although there are examples of similar companies following a somewhat similar track in the past, there is no real evidence to suggest that the directors have the expertise to manage this transition or to run the resulting company. Nor is it obvious that the resulting company is one in which the directors or the present employees would wish to work; certainly its character will be very different. In any case, their plans are only likely to be achieved if Syniad either seeks a stock exchange flotation or sells out to another company or other investors. The directors might possibly reconsider their plans and aim for a much more modest expansion, perhaps through agreements for collaboration with comparable companies elsewhere in Europe. However, it seems probable that Syniad has now reached a point where it can no longer thrive as a private company and its future must, inevitably, be very different from its past.